Products of Cosines: How the Entangling Layers of Quantum Classifier Heads Shape the CNN Representations Trained Through Them

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Abstract

When a convolutional network is trained end to end through a variational quantum circuit (VQC) classifier head, the head shapes the features the network learns, and we measure this with one common linear probe on the frozen bottleneck of each tuned network. On CIFAR-10 with 4,000 training images, an 8-qubit VQC head with CNOT rings lowers probe accuracy by 4.7 points relative to a linear head (5 seeds, every seed lower; a two-layer MLP probe finds the same gap), and removing the rings recovers 3.8 of those points. An exact identity organizes the pattern: with every variational rotation at zero, each readout is a product of cosines of the encoded inputs over a set fixed by the CNOT layers. Probe accuracy falls as the mean order of these products rises, from 73.6 at order 1 to 69.8 at the default order 2.5 and 60.1 at order 5.25; classical heads trained on the same products follow the same trend, the circuit ordering holds at twice the training budget and on MNIST, six new CNOT layouts predicted before training show a strong inverse association with order (two rank reversals), designed classical supports favor lower order at fixed mean overlap between readouts, and a circuit that retains entangling gates but has product order 1 recovers 3.1 points. On an ImageNet-pretrained ResNet-18 the same ordering holds: at an approximately matched number of optimizer updates the default head costs 6–8 points with 1,000 to 10,000 training images, while the ring-free head stays within 1.4 points of a linear head. Without CNOTs the head is exactly a classical per-coordinate sinusoidal model, and under matched per-head tuning a linear head has the highest mean test accuracy among linear, small-MLP and default VQC heads in every setting we test.

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

1. Introduction

A common near-term design for quantum image classification feeds a convolutional network into a small variational quantum circuit (VQC) that acts as the classifier head (Mari et al., 2020), building on variational quantum classifiers (Mitarai et al., 2018; Havlíček et al., 2019; Benedetti et al., 2019; Cerezo et al., 2021). When the network is trained jointly *through* the circuit, the head does more than classify: its gradients shape the features the backbone learns. A comparison between a quantum head and a classical one therefore compares two things at once, the heads and the representations they induce. The question reaches beyond quantum machine learning: a head that distorts the features trained through it is a known failure mode of fine-tuning (Kumar et al., 2022), and here the distortion turns out to track a structural quantity that can be computed before training. Benchmarks of quantum classifiers typically score accuracy on fixed inputs (e.g. Fulton et al., 2026; Slattery et al., 2022); Bowles et al. (2024) also replace the quantum layer of a trained dressed hybrid network with a classical one and visualize how the learned layers transform the data. Controls that isolate a circuit inside a hybrid model score predictive performance (Kölle et al., 2024; Elhag et al., 2026). To our knowledge, none of these measures quantitatively what the quantum head does to the representation trained through it, or which part of the circuit is responsible.

We answer both questions for a CNN with an 8-dimensional bottleneck and an 8-qubit head built from angle encoding $R_y(\pi \tanh z_j)$, four layers of trainable rotations each followed by a ring of CNOTs, and a linear readout of $\langle Z_k \rangle$. After joint training we freeze each backbone and fit the same linear probe to its bottleneck, so that probe accuracy compares the representations directly. The default VQC head lowers this probe accuracy, and the size of the drop follows the circuit's CNOT structure. An exact identity defines a structural descriptor whose order tracks this cost: when every trainable rotation is zero, which is near where training starts, each readout is $\langle Z_k \rangle = \prod_{j \in S_k} \cos(\pi \tanh z_j)$, a product of cosines over a set S_k of inputs fixed by the

CNOT layers alone (Figure 1a). The mean size of these sets, the *product order*, is 1 without CNOTs, 2.5 for the default circuit and 5.25 when a single ring follows the last layer.

Our contributions are:

1. **A tuning audit** (Section 4). Under a revised protocol (longer training, validation-based selection, each head's learning rate tuned), the VQC's apparent advantage over a near-parameter-matched MLP on MNIST shrinks from up to 18 points to a largest observed mean of 3.3. On CIFAR-10 the VQC still trails the MLP by 3.1–9.2 points, and among the linear, MLP and default VQC heads the linear head has the highest mean test accuracy in all six dataset–size settings.

2. **A representation cost tied to the CNOT layers** (Sections 5.1 and 5.2). On CIFAR-10 with 4,000 training images the default VQC head lowers bottleneck probe accuracy by 4.7 ± 0.9 points relative to a linear head (5 seeds), and a two-layer MLP probe finds the same gap, so the loss is not just one of linear readability. Removing the CNOT rings recovers 3.8 ± 1.2 points. On a pretrained ResNet-18 the default head costs 6–8 points at 1,000 to 10,000 images at an approximately matched number of updates (22.3 at 1,000 with a fixed 30 epochs); the ring-free head stays within 1.4 points of linear (Section 5.6). Five controls leave a substantial gap (Section 5.5).

3. **A structural account** (Sections 5.3 and 5.4). Probe accuracy falls with product order across eleven heads (Figure 1b). Classical heads trained on a circuit's zero-rotation products, with no circuit, follow the same trend; on the small CNN the default circuit's products match the VQC (-1.0 ± 2.2), and on the pretrained backbone they reproduce the ordering. Classical heads with designed supports favor lower order where overlap and rank are held fixed, though order alone does not set the cost, and the ordering replicates on MNIST and, under predictions recorded before the runs, on six new layouts (Spearman -0.89, two rank reversals). A circuit that keeps its CNOTs but has product order 1 recovers 3.1 ± 1.2 points over the default. Without CNOTs the head is exactly a classical per-coordinate sinusoidal model, and classical heads of that form come within 0.3–0.9 probe points of it.

2. Related Work

Benchmarking quantum classifiers and the role of entanglement. Large benchmarks find that simple classical models generally match or beat variational quantum classifiers, and that removing entanglement often does not hurt (Bowles et al., 2024); quantum fidelity kernels show a similar lack of advantage on classical data (Slattery et al., 2022). Closest to our setting, Fulton et al. (2026) show that classical $\mathrm{SU}(2)$ models match or exceed shallow VQCs on vision benchmarks and that ring entanglement can degrade accuracy on fixed features. Recent hybrid-network studies reach related conclusions with matched preprocessing and latent width (Freinberger & Moser, 2026) or matched width and parameter count (Park et al., 2026), and controlled benchmarks of quantum transfer learning find the outcome strongly configuration-dependent (Innan et al., 2026). Other work trains hybrid models end to end and studies the learned representation: Bowles et al. (2024) visualize the layers of a trained dressed hybrid network, Kashif & Shafique (2026) analyze expressibility and trainability under joint training, and quantum self-supervised learning evaluates frozen hybrid encoders with linear probes (Jaderberg et al., 2021; Edwards et al., 2026). We probe the supervised bottleneck *before* the circuit, the representation the circuit's gradients shape, and relate its accuracy to a quantity, product order, read off the CNOT pattern.

Isolating the circuit inside hybrid models. Kölle et al. (2024) separate classical and quantum contributions in hybrid architectures, building on sequential training that first fits a classical surrogate and then swaps in the circuit (Altmann et al., 2023). Elhag et al. (2026) delete, freeze or corrupt the circuit in quantum message passing and compare it with parameter-matched classical baselines and an exact classical surrogate for its bare R_y measurement, under a $\pi \tanh$ encoding. Encodings in hybrid convolutional models have also been compared (Monnet et al., 2024). These controls score predictive performance. We add a common probe on the jointly trained bottleneck and structural controls (Sections 5.3 and 5.5), none of which varies product order alone (the cancelling pairs also halve the CNOT count).

Fourier structure, classical surrogates and head-induced feature distortion. Angle-encoded circuits compute truncated Fourier series of their encoded angles (Schuld et al., 2021), including under data re-uploading (Pérez-Salinas et al., 2020), which classical models can represent and emulate (Schreiber et al., 2023); sinusoidal features also appear in classical networks as periodic activations and Fourier feature maps (Sitzmann et al., 2020; Tancik et al., 2020). Our zero-rotation identity is an elementary instance of this structure, used as an explanatory variable: it shows which products the CNOT layers create, and their order tracks a measured cost. In classical deep learning, Kumar et al. (2022) show that the head used during fine-tuning distorts pretrained features, and linear probes are a standard tool for reading what an intermediate layer encodes (Alain & Bengio, 2016). We apply the probe to the quantum-head setting, where the head is a fixed-form circuit rather than a

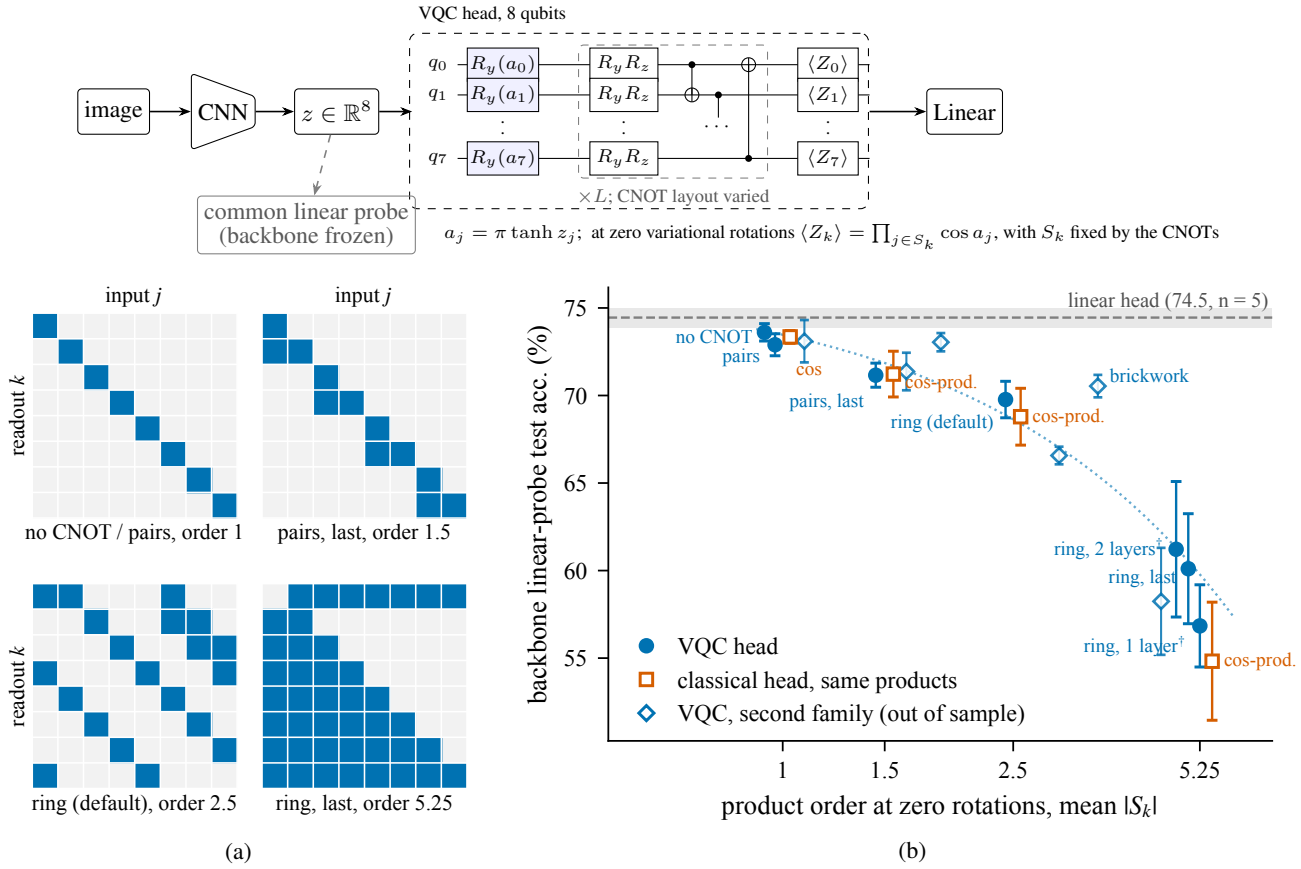


Figure 1. **(a)** A CNN feeds an 8-dimensional bottleneck z into an 8-qubit VQC head (default: four $R_y R_z$ layers, each followed by a CNOT ring $(q, q{+}1 \bmod 8)$; variants place CNOTs after the last layer only ("last"), use $L \in \{1, 2\}$ layers, or replace the ring by disjoint pairs $(0, 1), (2, 3), (4, 5), (6, 7)$). With every variational rotation at zero, each readout is exactly $\langle Z_k \rangle = \prod_{j \in S_k} \cos(\pi \tanh z_j)$; the grids mark $j \in S_k$ in blue (rows: readouts k, columns: inputs j). Pairs after every layer cancel at zero rotations, so their supports equal the no-CNOT ones. Product order is $\frac{1}{8} \sum_k |S_k|$. The backbone is evaluated by one common linear probe on the frozen bottleneck. **(b)** CIFAR-10 with 4,000 training images: probe test accuracy against product order (log axis) for VQC heads (filled) and classical heads trained end to end on the zero-rotation products of a circuit (open; at order 5.25 these are the supports of "ring, last" and "ring, 1 layer"; "ring, 2 layers" has different supports of the same mean order). Mean ± SD over 5 seeds (†: 3); head learning rate selected on mean validation accuracy over seeds 0–2, seeds 3–4 run at that choice. Dashed line and band: linear head. Points at one order are offset horizontally. The plot shows an association; order co-varies with support overlap. Open diamonds: six CNOT layouts outside the ring family (5 seeds, Section F), whose probe accuracy was predicted before training from the dotted fit over the ring-family (seeds 0–2) VQC circuits (linear in order, hence curved on the log axis); brickwork lies 5 points above its prediction.

learned classical layer.

Position of this work. That entanglement often does not help, and that trigonometric surrogates can stand in for simple circuits, are known. What is new here is (i) a measurement of the bottleneck a quantum classifier head induces under supervised joint training, with a probe shared across quantum and classical heads, and (ii) a structural variable, the order of the zero-rotation cosine products, that orders the representation cost across circuit and classical heads and survives an intervention that retains entangling gates while lowering order.

3. Setup

Model. A CNN backbone maps an image to a bottleneck $\mathbf{z} \in \mathbb{R}^8$ (MNIST: three convolutional blocks, about 94K parameters; CIFAR-10: four blocks, about 0.96M; Section 5.6 also uses an ImageNet-pretrained ResNet-18). Each head is trained jointly with its own copy of the backbone. The *linear* head is a single dense layer (90 parameters); *MLP-8* has one hidden layer of width 8 with ReLU (162). The default *VQC* head encodes $a_j = \pi \tanh z_j$ by $R_y(a_j)$ on qubit j, applies $L = 4$ layers of trainable $R_y R_z$ rotations on every qubit, each followed by a ring of CNOTs $(q, q{+}1 \bmod 8)$ applied in order $q = 0, \ldots, 7$, measures $\langle Z_k \rangle$ on every qubit, and maps the eight expectations to ten logits with a linear layer (154 stored parameters, 146 of which can affect the output).

Rotations are initialized in $\mathcal{U}(-0.1, 0.1)$. We simulate the circuit exactly with a batched PyTorch statevector simulator checked against PennyLane (Bergholm et al., 2018). Circuit variants change CNOT placement or rotation depth: *no CNOT*; a ring after the *last* layer only; $L \in \{1, 2\}$ layers; and disjoint *pairs* $(0, 1), (2, 3), (4, 5), (6, 7)$ after every layer or after the last layer only.

Protocol. For the small CNNs (the pretrained runs are specified in Section 5.6), training subsets are class-stratified samples of 250, 500 or 1,000 MNIST images and 1,000, 2,000 or 4,000 CIFAR-10 images; a disjoint stratified 2,000-image validation set selects the checkpoint epoch. Every run uses AdamW (weight decay 10^{-4}), a cosine schedule, batch size 32 and gradient clipping, for 50 epochs on MNIST and 80 on CIFAR-10 with standard augmentation. The backbone learning rate is half the head learning rate. For each dataset, size and head we select the head learning rate with the best mean validation accuracy over seeds 0–2 from a grid of half-decade steps, and report test accuracy once. Headline CIFAR-10/4,000 cells add seeds 3 and 4 at that selected rate. Heads in a cell share seeds, hence training subsets and validation splits, so we report paired differences (mean $\pm$ SD over seeds, in percentage points).

Common linear probe. To compare representations, we freeze each selected run's backbone, extract $\mathbf{z}$ on clean images, and fit a fresh linear layer $\mathbb{R}^8 \to \mathbb{R}^{10}$ (AdamW, cosine schedule, probe learning rate and epoch selected on validation). Its test accuracy, *probe accuracy*, compares the bottlenecks that the heads induced in joint training.

Zero-rotation identity. With every trainable rotation set to zero, the variational block reduces to its CNOTs. The encoded state is a product state with $\langle Z_j \rangle = \cos a_j$, and conjugating by $\mathrm{CNOT}(c, t)$ maps $Z_t \mapsto Z_c Z_t$ and leaves Z_c unchanged. Propagating Z_k back through the gates in reverse order, and cancelling repeated factors with $Z^2 = I$, gives a Pauli-Z string on a set S_k of qubits:

$$\langle Z_k \rangle\big|_{\theta=0} = \prod_{j \in S_k} \cos(\pi \tanh z_j), \qquad \bar{s} = \tfrac{1}{8} \textstyle\sum_k |S_k|. \tag{1}$$

We call $\bar{s}$ the *product order*. It is 1 without CNOTs, 2.5 for the default circuit, 5.25 for a ring after the last layer or a single-layer circuit, 1.5 for pairs after the last layer, and 1 for pairs after every layer: the pair layer P satisfies $P^2 = I$, so its four copies cancel at zero rotations, while interleaved nonzero rotations generally prevent the cancellation (Figure 1a; Section A lists the supports). Without CNOTs the identity extends to all rotation angles: each qubit evolves alone, so

$$\langle Z_j \rangle = \alpha_j \cos a_j + \beta_j \sin a_j, \tag{2}$$

with α_j, β_j set by qubit j's trained rotations. With its readout layer, the ring-free head is thus a classical per-coordinate sinusoidal model. To test whether the products themselves carry the effect, we train classical heads on them with no circuit: *cos* uses $\cos a_j$ (order 1), *cos+sin* adds $\sin a_j$, and *cos-prod.* uses the products of Equation (1) for a given circuit's supports, each followed by a linear layer and tuned like every other head.

4. Tuning Changes Head Comparisons, but a Linear Head Stays Ahead

Before turning to representations, we check how much a head comparison depends on the training recipe. In an exploratory sweep with one shared recipe (head learning rate 10^{-3} for every head, 25 MNIST epochs; Section I), the VQC beat MLP-8 on MNIST by 17.8, 18.1 and 6.9 points at 250, 500 and 1,000 images and lost to it on CIFAR-10 by 14.2, 13.8 and 8.3 points. Under the protocol of Section 3 (50 MNIST epochs, validation-based selection, per-head learning rates), these differences change in both size and sign (Table 1). On MNIST, VQC $-$ MLP-8 becomes -2.1 ± 9.7, $+3.3 \pm 2.6$ and $+0.8 \pm 1.0$. Part of this change comes from the protocol itself: at the shared rate under the new protocol the advantage is already $+10.8$, $+6.1$ and -0.5 (last column of Table 1), and per-head tuning moves it to the values above. On CIFAR-10, where only the learning rate changes between the two columns of Table 1, the VQC deficit to MLP-8 shrinks from 10.1–13.6 points to 3.1–9.2, and the VQC's own test accuracy rises by 6.0, 6.1 and 10.7 points. A linear head has the highest mean in all six settings and beats the VQC for every seed; its mean lead is 1.7–13.7 points on MNIST and 7.0–15.8 on CIFAR-10. Every selected rate is interior to its grid; for the linear head this took two extension steps in five settings.

This audit fixes the comparisons used in the rest of the paper: every head is evaluated at its own validated learning rate. It also leaves a question. At 4,000 CIFAR-10 images the VQC trails the linear head by 7.0 ± 1.2 test points (5 seeds: 6.7 ± 1.0). Part of that gap could come from the VQC as a readout and part from the backbone it trains; a common probe measures the gap that remains under one readout.

5. What the VQC Head Does to the Backbone

5.1. The default VQC head lowers bottleneck probe accuracy

On CIFAR-10 with 4,000 images, the common probe reaches 69.8 ± 1.0 on backbones trained with the default VQC head and 74.5 ± 0.5 on backbones trained with a linear head, a paired difference of -4.7 ± 0.9 points that is negative for all five seeds. The probe reads the VQC-trained bottleneck better than the VQC head itself does (67.8 ± 1.3

Table 1. Test accuracy (%) of three heads on the same CNN, each with its head learning rate selected by mean validation accuracy over the tested grid (backbone lr is coupled at $0.5\times$ the head lr); mean $\pm$ SD over 3 seeds; bold: highest mean in the row. MLP-8: one hidden layer of width 8; VQC: the default 8-qubit, 4-layer ring circuit. Differences are in percentage points, paired over seeds (same training subset and split). The last column repeats VQC $-$ MLP-8 with both heads at one shared head lr (10^{-3}). Every selected lr is interior to its grid.

		Selected lr			VQC $-$ (selected lr)		VQC $-$ MLP-8
Data	N_{train}	Linear	MLP-8	VQC	Linear	MLP-8	(lr 10^{-3})
MNIST	250	$\mathbf{90.3 \pm 1.9}$	78.7 ± 6.5	76.6 ± 5.3	-13.7 ± 5.9	-2.1 ± 9.7	$+10.8 \pm 13.9$
	500	$\mathbf{94.3 \pm 0.2}$	86.7 ± 3.0	90.0 ± 0.6	-4.3 ± 0.5	$+3.3 \pm 2.6$	$+6.1 \pm 11.6$
	1000	$\mathbf{96.4 \pm 0.5}$	93.9 ± 0.8	94.6 ± 0.4	-1.7 ± 0.9	$+0.8 \pm 1.0$	-0.5 ± 3.8
CIFAR-10	1000	$\mathbf{59.0 \pm 0.9}$	52.3 ± 1.8	43.1 ± 3.8	-15.8 ± 3.8	-9.2 ± 4.5	-13.6 ± 1.8
	2000	$\mathbf{67.6 \pm 0.7}$	62.0 ± 1.0	53.1 ± 0.9	-14.5 ± 1.3	-8.9 ± 1.7	-13.5 ± 2.0
	4000	$\mathbf{74.6 \pm 0.5}$	70.8 ± 0.5	67.7 ± 1.1	-7.0 ± 1.2	-3.1 ± 0.9	-10.1 ± 1.6

test): replacing the VQC head by the probe raises accuracy by about 2.0 points, and a 4.7-point gap remains between the two backbones under the common probe. The gap is larger with less data (CIFAR-10: -11.2 ± 4.0 and -11.1 ± 1.5 at 1,000 and 2,000 images) and smaller on MNIST (-7.9 ± 4.2, -2.2 ± 0.8 and -1.0 ± 0.5 at 250, 500 and 1,000; 3 seeds each). A narrow classical head lowers probe accuracy as well: MLP-8 backbones probe 3.6 ± 0.6 points below linear-head backbones on CIFAR-10/4,000, 1.0 ± 0.9 above the VQC. The gap is not an artifact of linear decoding: a two-layer MLP probe on the same bottlenecks recovers at most 0.4 points more than the linear probe for any head, and puts the default VQC 4.8 ± 0.9 points below the linear head (Section E), also when its learning-rate grid is extended downward. The gap therefore persists under the tested linear and two-layer MLP probes; we do not claim more about the information the bottleneck contains. The rest of this section shows that the VQC's share of the gap depends on its CNOT structure, which no classical head has.

5.2. Removing the CNOT rings recovers most of the gap

Deleting every CNOT and keeping the rest of the head unchanged raises probe accuracy to 73.6 ± 0.5 and test accuracy to 73.1 ± 0.5 (5 seeds). Against the default VQC this is $+3.8 \pm 1.2$ probe and $+5.2 \pm 1.4$ test points, positive for every seed, and the ring-free head now probes 2.8 ± 0.6 points above MLP-8. It leaves 0.8 ± 0.5 probe and 1.4 ± 0.5 test points to the linear head. MNIST shows the same direction: ring removal gains 2.8 ± 0.4 and 1.4 ± 0.4 probe points at 500 and 1,000 images, and the ring-free head's test accuracy (94.5, 96.4) equals the linear head's (94.3, 96.4) to within 0.2 points.

5.3. Product order tracks the gap

Which property of the rings matters? Figure 1b places every head at the product order of Equation (1). Across the eleven ring-family heads, at four orders, every head's mean probe accuracy is below that of every lower-order head. Depth and parameter count do not explain this: a ring after the last layer keeps the default's parameter count and rotation layers but raises the order to 5.25 and lowers the probe by 9.7 ± 2.2 points (5 seeds), and the 1- and 2-layer circuits at that order probe at 56.8 and 61.2 (3 seeds).

Order also carries over to layouts it was not fit on. For six CNOT layouts outside the ring family, chosen by order alone, we predicted probe accuracy from the ring-family fit before training. The association holds (Spearman -0.89, against a threshold of -0.8 set in advance; five of six within 4 points), with two rank reversals. The largest miss, brickwork at 5 points above its prediction, is the layout trained furthest from its product model; at its trained rotations its effective order is 2.6 rather than 3.5 (Figure 1b, Sections F and G). Order is a useful but coarse descriptor.

Two interventions test the products directly. First, classical heads trained end to end on a circuit's zero-rotation products, with no circuit, land at the circuit's order on the trend: the products of the default circuit give 68.8 ± 1.6 probe and 67.1 ± 1.3 test, paired differences to the default VQC of -1.0 ± 2.2 and -0.7 ± 1.7 with mixed signs across seeds. Summing instead of multiplying the same cosines would give the order-1 cos head's function class (73.3 probe), as every CNOT-derived support matrix has full rank, though such a head would train differently. Second, the *pairs* circuit keeps a CNOT after every layer but has order 1, because its pair CNOTs cancel at zero rotations. It recovers 3.1 ± 1.2 probe and 4.4 ± 1.2 test points over the default, positive for every seed, and sits 0.7 ± 1.0 probe points below the ring-free circuit. Pairs after the last layer only (order 1.5) probe at 71.2 ± 0.7, between orders 1 and 2.5, as does the classical head on their products (71.2 ± 1.3).

In CNOT-derived supports, order rises with overlap, the number of inputs two readouts share, so classical product heads on designed supports separate the two (Table 7). At equal overlap and full rank, the order-3 star probes 5.4 ± 2.7 points above an order-4 layout (5 seeds, all positive); the other matched comparisons point the same way but also

change rank or match overlap only approximately. At equal order the stars, which reuse one or two inputs across most products, probe 3.0 and 1.8 points above the windows (the latter with mixed signs), so how the coupling is distributed also matters, and an order-3 star probes as high as the default ring's order-2.5 products (68.9 vs 68.8). The designed heads also differ in training fit (85.6% vs 74.1% clean training accuracy for the order-3 star and the order-4 layout), so their contrasts mix structure with optimization. Within the tested family and protocols, higher order tracks lower mean probe accuracy, but order is not sufficient to determine it.

The pattern also replicates on MNIST with 1,000 images, where we fixed the layouts and recorded the predicted ordering before running them. Probe accuracy is 96.5 without CNOTs, 95.5 for pairs after the last layer, 95.1 for the default circuit and 91.6 for a ring after the last layer; the small pairs-last advantage over the default is $+0.4 \pm 0.5$ (5 seeds, 4 positive); the classical head on the default products probes 0.1 ± 0.5 below the default VQC, and the one on the last-ring products falls below that circuit (90.4).

The identity is exact only at zero rotations, so we fit an affine map from each trained circuit's zero-rotation readouts to its trained readouts on half of the validation set and evaluate it on the other half (Section B; a frozen circuit gives $R^2 = 0.997$). For the default and last-ring circuits it recovers 87% and 90% of the held-out readout variance, and the bottleneck gradient through the remapped products aligns with the trained one (median cosine 0.86 and 0.87). The products miss some task detail: the trained readout on the remapped products scores 7 and 6 points lower. Low-order circuits move further (R^2 0.73–0.83; for the ring-free circuit through the $\sin a_j$ term of Equation (2)). At order 5.25 the trained rotations may recover part of the cost: the last-ring circuit tests 6.4 ± 5.8 points above the classical head on its products (four of five seeds; 95% interval $[-0.8, 13.6]$). The cost does not depend on starting in the product regime: with the rotations initialized in $\mathcal{U}(-\pi, \pi)$, where the identity does not hold at the start, the default circuit still probes 4.6 ± 1.7 points below the ring-free one (3.8 ± 1.6 on the same three seeds at the default initialization), and neither circuit's probe moves by more than a point on average (Table 12); there each default readout still depends on all eight inputs (Section G).

5.4. Without CNOTs the head is a classical sinusoidal model

By Equation (2), the ring-free head computes $W[\alpha_j \cos a_j + \beta_j \sin a_j]_j + b$ for every setting of its rotations. Table 2 compares it with classical heads of this form trained under the same protocol. The cos-only restriction trails it by 0.3 ± 0.6 probe and 0.6 ± 0.7 test points with mixed signs, and the unrestricted cos+sin extension by 0.9 and 1.3 (3 seeds). The match holds at the level of the function class, not the parameters: the trained ring-free circuit has mean absolute R_y angle 0.41 rad and uses the sine component (Jacobian cosine 0.41 against the cos-only model; Table 6).

5.5. Controls that do not recover the gap

Table 3 tests alternative explanations on CIFAR-10/4,000. Lowering or raising the backbone learning rate with the head learning rate fixed leaves a gap of 4.9–8.6 points at every rate where the VQC trains, and the VQC collapses at the two highest rates. The backbone's gradient norm is larger, not smaller, with the VQC head. Freezing the circuit's rotations at initialization, which keeps the head close to its zero-rotation model, does not recover the probe (-1.6 ± 0.7 against the trained circuit). BatchNorm before the encoding and two alternative encodings, one of which breaks the sign symmetry of $\cos(\pi \tanh z)$, recover at most 0.7 probe points. Lowering product order gives by far the largest recoveries.

Higher-order circuits also fit their training data less well under the 80-epoch budget (clean training accuracy 96.6% without CNOTs, 89.2% for the default circuit and 65.6% for a ring after the last layer), so part of the gap could be under-training. Doubling the budget to 160 epochs for the linear, default, ring-free and last-ring heads (CIFAR-10/4,000) raises every probe but keeps the ordering (Table 8). The default circuit then fits its training set nearly as well as the ring-free and linear heads (98.8% vs 99.9%) and still probes 2.8 ± 0.4 points below the ring-free head and 3.1 ± 0.3 below the linear one (5 seeds, every seed negative). The last-ring circuit remains under-fit at 160 epochs (79.5%), so its larger cost includes slower optimization.

5.6. A pretrained backbone, data size and training budget

Hybrid image classifiers are often built on a pretrained network. We repeat the key heads on an ImageNet-pretrained ResNet-18 fine-tuned through each head, its classifier replaced by the 8-dimensional bottleneck (Table 4). The ordering carries over: at 4,000 images the ring-free and pairs circuits probe within 0.2 points of the linear head, the default circuit 7.2 ± 2.3 below it and the last-ring circuit 18.0 ± 4.7 below (5 seeds, every seed negative); on the three seeds probed with both, an MLP probe gives the same default gap (-5.8 vs -5.7). At a fixed 30 epochs the cost falls with training size, from 22.3 ± 2.9 points at 1,000 images to 1.9 ± 0.4 at 10,000, but smaller sets then also get fewer optimizer updates. At an approximately matched number of updates (about 3,800) the default circuit costs 6.6 ± 1.7, 5.7 ± 0.7 and 7.2 ± 2.1 points at 1,000, 4,000 and 10,000 images, and the ring-free head stays within 0.5 points of linear (seeds 0–2; Table 15); re-selecting the learning rates

Table 2. Without CNOT rings the VQC head is exactly a classical per-coordinate sinusoidal model, $\text{logits} = W[\alpha_j \cos a_j + \beta_j \sin a_j]_{j=1}^{8} + b$ with $a_j = \pi \tanh z_j$ and α_j, β_j set by the trained rotations, for all rotation angles. Classical heads $W[\cos a_j]_j + b$ (a restriction) and $W[\cos a_j, \sin a_j]_j + b$ (an unrestricted extension), trained end to end with the same recipe and per-head validation lr selection, trail it by 0.3–0.9 probe points and 0.6–1.3 test points. CIFAR-10 with 4,000 training images; test accuracy of the full model and of the common linear probe on the bottleneck (%, mean ± SD over n seeds); differences in percentage points, paired over seeds.

				− ring-free VQC	
Head	n	Test	Probe	Test	Probe
VQC, no CNOT rings	5	73.1 ± 0.5	73.6 ± 0.5	—	—
cos-only (restriction)	5	72.5 ± 0.5	73.3 ± 0.4	-0.6 ± 0.7	-0.3 ± 0.6
cos+sin (unrestricted extension)[a]	3	72.0 ± 0.2	72.8 ± 0.4	-1.3 ± 0.4	-0.9 ± 0.5
linear (reference)	5	74.5 ± 0.7	74.5 ± 0.5	$+1.4 \pm 0.5$	$+0.8 \pm 0.5$

[a] 3 seeds; differences are against the ring-free VQC on the same seeds (test 73.3, probe 73.7).

Table 3. Controls on CIFAR-10 with 4,000 training images. Entries are differences in the test accuracy of the common linear probe on the frozen 8-dimensional bottleneck, in percentage points, paired over 3 seeds (mean ± SD). The five controls leave a substantial probe gap; removing the CNOT rings recovers most of it.

Control	Comparison	Probe Δ	Observation
Reference: default VQC	VQC − linear	-4.7 ± 1.2	the gap under study
Backbone lr[a]	VQC − linear, backbone lr 0.0025	-6.3 ± 1.2	gap persists at every tested backbone lr where the VQC trains
	VQC − linear, backbone lr 0.005	-4.9 ± 1.2	
	VQC − linear, backbone lr 0.015	-8.6 ± 0.5	
Freeze circuit at init	frozen − trained circuit	-1.6 ± 0.7	gap persists with rotations frozen at initialization
BatchNorm before encoding	VQC+BN − VQC	-1.2 ± 2.2	pre-encoding BN does not recover the probe
	VQC+BN − linear	-5.9 ± 3.4	
Monotone encoding[b]	monotone − default	$+0.3 \pm 0.7$	neither encoding recovers more than 0.7 points
	symmetric control − default	$+0.7 \pm 0.5$	
	monotone − symmetric control	-0.4 ± 0.9	
Contrast: remove CNOT rings	ring-free − default	$+3.8 \pm 1.6$	recovers most of the gap (§5.2)
	ring-free − linear	-0.8 ± 0.7	

[a] Head lr fixed at 0.01 (VQC) and 0.03 (linear) while the backbone lr varies; the VQC collapses at the tested backbone lrs 0.03 and 0.06. Other rows: each head at its validation-selected head lr, backbone lr $0.5\times$ head lr. [b] Monotone: $R_y(\frac{\pi}{2}\tanh z + \frac{\pi}{2})$; symmetric control: $R_y(\frac{\pi}{2}\tanh z)$, same range width; default: $R_y(\pi \tanh z)$. Not a probe difference: the final-epoch pre-clip backbone gradient norm is larger with the VQC head than with the linear head (1.92 ± 0.13 vs 0.75 ± 0.02).

gives 8.3 ± 2.1 and 7.7 ± 1.9 at 1,000 and 10,000 images, with the ring-free head 1.4 ± 1.2 and 0.7 ± 0.2 below linear. Training budget therefore accounts for much of the size dependence at fixed epochs; with one matched budget and a schedule that changes with the epoch count, data size and budget are not fully separated.

Here the high-order heads are also unstable across runs (test accuracy spans up to 27 points over three seeds at one learning rate). In the main grid the classical head on the default circuit's products probes above the circuit (-4.0 ± 1.6 vs -7.2 ± 2.3 against linear, every seed), each at its validation-selected rate. When the circuit, the same circuit frozen at zero rotations and the products head are trained in lockstep, from copied weights on identical batches (Section H), they do not differ detectably at a shared learning rate (circuit − products -0.6 ± 2.0 and $+0.6 \pm 3.9$ at 3×10^{-4} and 10^{-3}, 10 seeds; 95% intervals $[-2.0, 0.8]$ and $[-2.1, 3.4]$), while at the main-grid rates the products head stays ahead ($+2.9 \pm 3.0$, 8 of 10 seeds). Learning-rate selection is thus the more likely source of the products head's advantage, without excluding a contribution of the rotations.

6. Discussion and Limitations

A mechanism hypothesis. Why should product order cost the backbone? Differentiating Equation (1) at zero rotations

Table 4. The representation cost on an ImageNet-pretrained ResNet-18 fine-tuned through each head (CIFAR-10; backbone learning rate $0.1\times$ the head's; 30 epochs at 128 px). Probe: test accuracy (%) of the common linear probe on the frozen 8-dimensional bottleneck, and its paired difference to the linear head in percentage points (mean $\pm$ SD; 4,000 images: number of seeds in parentheses; 1,000 and 10,000 images: 3 seeds). Head learning rates are selected on validation separately at each size. At 4,000 images every head's grid spans 3×10^{-4}–10^{-2}, extended to 10^{-4} for the default and last-ring circuits; at 10,000 images it spans 3×10^{-4}–3×10^{-3}, extended to 10^{-4} for the linear, default and last-ring heads. At both sizes every selected rate is interior. The linear row gives its absolute probe accuracy at 1,000 and 10,000 images. The last column repeats the 4,000-image difference with a two-layer MLP probe (3 seeds).

		1,000 images	4,000 images		10,000 images	4,000, MLP probe
Head	Order	− linear	probe	− linear	− linear	− linear
linear	—	82.9 ± 0.9	89.4 ± 0.4 (5)	—	92.6 ± 0.4	—
VQC, no CNOT	1	-1.6 ± 1.2	89.4 ± 0.4 (5)	0.0 ± 0.1	-0.1 ± 0.4	-0.6 ± 0.3
VQC, pairs	1		89.3 ± 0.4 (5)	-0.2 ± 0.3		-0.8 ± 0.5
cos-prod. (default supports)	2.5		85.5 ± 1.6 (5)	-4.0 ± 1.6		-3.0 ± 1.1
VQC, ring (default)	2.5	-22.3 ± 2.9	82.3 ± 2.3 (5)	-7.2 ± 2.3	-1.9 ± 0.4	-5.8 ± 0.3
VQC, ring, last	5.25	-34.1 ± 6.3	71.4 ± 4.8 (5)	-18.0 ± 4.7	-8.0 ± 1.0	-18.5 ± 3.9

gives, for $j \in S_k$,

$$\frac{\partial \langle Z_k \rangle}{\partial z_j} = -\pi \operatorname{sech}^2(z_j) \sin a_j \prod_{i \in S_k \setminus \{j\}} \cos a_i. \quad (3)$$

The gradient a bottleneck coordinate receives through readout k is scaled by the cosines of every other coordinate in S_k, and it approaches zero when any of them nears $a_i = \pm\pi/2$. For independent coordinates its mean square has an upper bound that falls geometrically with the support size when $\mathbb{E}[\cos^2 a_i] < 1$ (Proposition 1; measured factor 0.6–0.8 per input). We hypothesize that this multiplicative coupling, absent at order 1, makes the class signal each coordinate receives depend on the others, so that higher-order heads train bottlenecks with less probe-recoverable class signal. Consistent with this, the effective rank (of 8) of the per-sample bottleneck gradients $\partial \mathcal{L}/\partial z$ falls from 7.2 for the ring-free head (7.3 for linear) to 4.5 for the default and 3.7 for the last-ring circuit, ring-free above default in every seed, with the same mean ordering at initialization (Table 11); this is a correlate, not a proof of cause; on the second family it ranks the layouts as their probe accuracy does, but misses most of brickwork's advantage (Section F). The MLP probe argues against the signal being present but only nonlinearly readable (Section E). Removing the coupling directly, by passing each input's gradient only through its own factor, broke optimization (55.7% vs 84.0% clean training accuracy; Table 10) and is inconclusive, so the coupling account remains a hypothesis.

Implications for hybrid design. For CNN-fed quantum heads trained end to end, the CNOT layout is a choice about the backbone: product order, computable before training, ranks the ring-family circuits on both datasets and is strongly associated with the cost of new layouts out of sample, with two rank reversals. The best quantum heads are low-order, and the best, the ring-free head, is exactly classical (Equation (2)). Head comparisons should tune each head separately and measure the backbone, not only accuracy.

Limitations. We study two families of CNOT layouts (8 qubits, $R_y R_z$ layers, $\pi \tanh$ angle encoding with two variants) on an 8-dimensional bottleneck, one qubit per coordinate, in noiseless simulation, with small CNNs and one pretrained ResNet-18 on subsets of MNIST and CIFAR-10 of at most 10,000 images; the size of the effect with other encodings, other pretrained models or on hardware is untested. Most cells use 3 seeds (5 or 10 where stated), enough for consistent signs in paired comparisons but not for equivalence claims. Product order is a coarse scalar: it misses brickwork by 5 points, and at fixed order the distribution of coupling matters by an amount we do not pin down (Table 7); the designed supports were tested with classical heads only. The identity is exact only at zero rotations, and trained circuits on the small CNNs move away from it (held-out R^2 0.51–0.90). Finally, our probes are linear and a two-layer MLP; information recoverable only by stronger decoders is not measured.

7. Conclusion

A quantum classifier head trained jointly with a CNN changes the representation the CNN learns. With every head tuned on validation data and every selected backbone read by the same linear probe, the default ring-entangled VQC head lowers CIFAR-10 probe accuracy by 4.7 points, and the drop tracks the order of the cosine products each readout computes at zero rotations, also on new layouts predicted before training. Lowering that order, by removing the rings or by entangling only in pairs that cancel at zero rotations, recovers most of the gap, and classical heads on the same products follow it. On a pretrained ResNet-18 the default head costs 6–8 points at every training size when updates are approximately matched. Without CNOTs the head is a classical sinusoidal model. In practice: tune each head separately, probe the backbone, and count a circuit's product order before training through it.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning, specifically the evaluation of hybrid quantum–classical classifiers. Its practical consequence is methodological: comparisons between quantum and classical heads should tune each head separately and measure the representation the head induces. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. The Zero-Rotation Identity

Let $U(\theta)$ be the variational block and $|\psi(\mathbf{z})\rangle = \bigotimes_j R_y(a_j)\,|0\rangle$ the encoded product state, so that $\langle Z_k\rangle = \langle\psi|\,U^\dagger Z_k U\,|\psi\rangle$. At $\theta = 0$ every R_y and R_z is the identity and U is the product of the circuit's CNOTs. Conjugation by $\mathrm{CNOT}(c,t)$ maps $Z_t \mapsto Z_c Z_t$ and fixes Z_c, so $U^\dagger Z_k U$ is obtained by applying these rules to Z_k for the gates in reverse order; the result is a string $\prod_{j\in S_k} Z_j$, with S_k the set of qubits on which an odd number of propagation paths land. Since $|\psi\rangle$ is a product state with $\langle\psi|\,Z_j\,|\psi\rangle = \cos a_j$, the expectation factorizes into Equation (1). Table 5 lists the supports. Pairs after every layer apply each pair CNOT four times, which is the identity, so their supports equal the no-CNOT ones; with trained rotations between the layers the gates no longer cancel (its outputs differ from the ring-free circuit's at random rotation angles), and the circuit can entangle: with every input at zero and every rotation zero except $R_y(\pi/2)$ on qubit 0 in the last layer, the final pair CNOT turns qubits 0 and 1 into a Bell pair, whose single-qubit reduced state has purity $1/2$.

For the ring-free circuit and any rotation angles, qubit j starts from Bloch vector $(\sin a_j, 0, \cos a_j)$ after encoding and then undergoes a fixed rotation $M_j \in \mathrm{SO}(3)$ determined by its trained $R_y R_z$ layers. Hence $\langle Z_j\rangle = (M_j)_{zx}\sin a_j + (M_j)_{zz}\cos a_j$, which is Equation (2) with $\beta_j = (M_j)_{zx}$ and $\alpha_j = (M_j)_{zz}$. The test suite checks the formula against the simulator (tolerance 10^{-10}). Since the readout layer is linear, the ring-free head is $W[\alpha_j\cos a_j + \beta_j \sin a_j]_j + b$.

Table 5. Zero-rotation supports S_k of each readout $\langle Z_k\rangle$, written as digit strings (e.g. 015 is $\{0,1,5\}$), and the mean product order $\bar{s}$, computed by propagating Z_k backwards through each circuit's CNOTs.

Circuit	$\bar{s}$	S_0	S_1	S_2	S_3	S_4	S_5	S_6	S_7
no CNOT	1	0	1	2	3	4	5	6	7
pairs (every layer)	1	0	1	2	3	4	5	6	7
pairs, last	1.5	0	01	2	23	4	45	6	67
ring (default)	2.5	015	256	367	047	15	26	37	04
ring, last	5.25	1234567	01	012	0123	01234	012345	0123456	01234567
ring, $L = 1$	5.25	1234567	01	012	0123	01234	012345	0123456	01234567
ring, $L = 2$	5.25	01357	0234567	134567	024567	13567	02467	1357	0246

B. Trained Circuits against the Product Model

Table 6. Trained circuits against their own zero-rotation product model (CIFAR-10, 4,000 training images; small CNN, and below the pretrained ResNet-18; each run's validation embeddings, split by index parity; means over n seeds). An affine map from the readouts of the same circuit with all rotations set to zero, q_0, to the trained readouts q is fit on one half and evaluated on the other. R^2: held-out fraction of trained readout variance it recovers, a ratio relative to the circuit's own output spread; MSE: the unnormalized held-out mean squared error per readout ($\langle Z\rangle \in [-1,1]$). Accuracy: held-out accuracy of the trained head, and of its trained readout layer applied to the remapped q_0. Loss-gradient cosine: median over held-out samples of the cosine between $\partial\mathcal{L}/\partial z$ (cross-entropy) of the trained head and of the zeroed head with the trained readout kept (zeroed) or applied to the remapped q_0 (remapped). Last column: mean $|\theta|$ of the R_y / R_z rotations (rad).

					Accuracy		Loss-grad. cosine		
Circuit	Order	n	R^2	MSE	trained	remapped	zeroed	remapped	$\lvert R_y\rvert$ / $\lvert R_z\rvert$
ring (default), frozen at init	2.5	3	0.997	0.0010	67.6	67.0	1.00	1.00	0.05 / 0.05
ring (default)	2.5	5	0.866	0.0414	69.3	62.3	0.59	0.86	0.14 / 0.48
ring, last	5.25	5	0.903	0.0326	55.7	49.4	0.75	0.87	0.10 / 0.41
pairs, last	1.5	5	0.826	0.0684	70.0	60.4	0.49	0.81	0.20 / 0.37
pairs	1	5	0.788	0.0990	73.9	64.3	0.42	0.79	0.32 / 0.43
no CNOT	1	5	0.734	0.1309	74.1	61.9	0.34	0.71	0.41 / 0.28
Pretrained ResNet-18 backbone									
ring (default)	2.5	5	0.997	0.0019	78.2	78.1	0.98	0.98	0.02 / 0.05
ring, last	5.25	5	0.994	0.0037	58.6	59.3	0.91	0.91	0.05 / 0.05
no CNOT	1	5	0.992	0.0067	90.4	90.2	0.93	0.93	0.06 / 0.05

Table 6 compares each trained circuit with the same circuit after setting every rotation to zero, on the validation embeddings of the run's own backbone. The frozen circuit anchors the measurement: rotations held at their $\mathcal{U}(-0.1, 0.1)$ initialization reproduce the product model almost exactly. Held-out R^2 equals the in-sample value to within 0.015 for every run, so the affine map does not overfit. These are endpoint measurements: they say how close the trained circuit is to its product model,

not that training followed the product model's trajectory. Two results bound the approximation. First, on the small CNN the products, even after remapping, lose task-relevant detail: held-out accuracy drops by 6–12 points (by under 1 point on the pretrained backbone). Second, gradient alignment depends on the remapping: with the trained readout applied directly to the zeroed circuit, the median loss-gradient cosine is 0.59 for the default circuit, and -0.08 for its seed 4 (0.96 after remapping); after remapping it is 0.79–0.87 for every entangling small-CNN circuit (0.91–0.98 on the pretrained backbone), with 3–7% of samples negative on average (18% in the worst run). Zeroing the rotations without remapping costs 14–23 accuracy points on average for the two ring circuits and 31–43 for the pairs and CNOT-free circuits, which mostly reflects the unrefitted readout. Low-order circuits drift further, as Section 5.4 explains for the ring-free case.

C. Order versus Overlap, and Training Budget

Table 7. Classical product heads whose supports separate product order from overlap (mean pairwise $|S_k \cap S_l|$), trained end to end like every other head (CIFAR-10, 4,000 training images; n seeds, each at its validation-selected learning rate; differences paired over shared seeds). Rank: rank of the 8×8 support matrix, which bounds the number of independent input directions the products can respond to. Window w: $S_k = \{k, \ldots, k{+}w{-}1\}$; gapped window 4: $\{k, k{+}1, k{+}2, k{+}4\}$ (mod 8). Star 2: $\{1,2\}, \{0,1\}, \{0,2\}, \ldots, \{0,7\}$. Star 3: $\{1,2,3\}, \{0,1,2\}, \{0,1,3\}, \ldots, \{0,1,7\}, \{0,2,3\}$. Bottom: paired probe differences and what each comparison holds fixed.

Supports	Order	Overlap	Rank	n	Test	Probe
cos (order 1)	1	0.00	8	3	72.2 ± 0.3	73.2 ± 0.1
window 2	2	0.29	7	3	68.1 ± 0.4	69.3 ± 0.4
star 2	2	0.82	8	5	69.8 ± 2.2	70.9 ± 1.9
default ring products	2.5	0.57	8	3	66.3 ± 0.9	67.8 ± 1.1
window 3	3	0.86	8	5	64.2 ± 2.9	67.1 ± 2.3
star 3	3	1.71	8	5	67.2 ± 1.2	68.9 ± 1.1
window 4	4	1.71	5	5	60.1 ± 3.9	63.3 ± 3.1
gapped window 4	4	1.71	8	5	61.0 ± 2.4	63.5 ± 1.8

Comparison	Held fixed	Probe difference
star 3 − gapped window 4	equal overlap, both rank 8	$+5.4 \pm 2.7$
star 2 − window 3	approximately equal overlap, both rank 8	$+3.9 \pm 1.9$
star 3 − window 4	equal overlap, rank 8 vs 5	$+5.6 \pm 2.2$
star 2 − window 2	equal order, rank 8 vs 7	$+3.0 \pm 0.2$
star 3 − window 3	equal order, both rank 8	$+1.8 \pm 2.8$

Table 8. Twice the training budget (160 instead of 80 epochs, each head at its 80-epoch learning rate and cosine schedule; CIFAR-10, 4,000 training images; n seeds). Clean-train: accuracy on the unaugmented training subset at the selected checkpoint; loss: final-epoch training loss on augmented batches. At 160 epochs, default − no-CNOT probe accuracy is -2.8 ± 0.4 and default − linear is -3.1 ± 0.3 (paired).

Head	Epochs	n	Clean-train	Train loss	Test	Probe
linear	80	5	98.7 ± 0.4	0.13	74.5 ± 0.7	74.5 ± 0.5
	160	5	99.9 ± 0.1	0.03	75.6 ± 0.6	75.5 ± 0.4
no CNOT	80	5	96.6 ± 0.7	0.22	73.1 ± 0.5	73.6 ± 0.5
	160	5	99.9 ± 0.1	0.03	74.6 ± 0.5	75.1 ± 0.4
ring (default)	80	5	89.2 ± 2.5	0.44	67.8 ± 1.3	69.8 ± 1.0
	160	5	98.8 ± 0.6	0.08	70.6 ± 0.5	72.4 ± 0.6
ring, last	80	5	65.6 ± 5.4	1.02	55.3 ± 3.1	60.1 ± 3.1
	160	3	79.5 ± 15.1	0.67	59.6 ± 7.6	64.7 ± 4.7

Table 7 lists the designed-support heads of Section 5.3. Overlap is the mean over ordered readout pairs $k \neq l$ of $|S_k \cap S_l|$; if input j appears in d_j supports, order is $\frac{1}{8}\sum_j d_j$ and overlap is $\frac{1}{56}\sum_j d_j(d_j - 1)$, so for windows, where every d_j equals the order, the two rise together. Stars hold the order while concentrating the d_j on one or two inputs. Windows of width 2 and 4 have rank-deficient support matrices; the gapped window of width 4 restores full rank at the same order and overlap and probes 0.2 ± 4.1 points above the rank-5 window (mixed signs), so restoring full rank does not remove the order-4 deficit. Table 8 gives the training-budget check of Section 5.5, with the linear head also rerun at 160 epochs.

D. Gradient Attenuation at Zero Rotations

Proposition 1. *At zero rotations, for $j \in S_k$ and $a_i = \pi \tanh z_i$,*

$$\Big|\frac{\partial\langle Z_k\rangle}{\partial z_j}\Big| \le \pi \prod_{i\in S_k\setminus\{j\}} |\cos a_i|.$$

If, in addition, the coordinates z_i, $i \in S_k$, are independent, then

$$\mathbb{E}\Big[\Big(\frac{\partial\langle Z_k\rangle}{\partial z_j}\Big)^2\Big] \le \pi^2 \prod_{i\in S_k\setminus\{j\}} \mathbb{E}\big[\cos^2 a_i\big] \le \pi^2\rho^{|S_k|-1}, \qquad \rho = \max_{i\in S_k} \mathbb{E}\big[\cos^2 a_i\big] \le 1.$$

Proof. Differentiating Equation (1) gives $|\partial\langle Z_k\rangle/\partial z_j| = \pi\,\mathrm{sech}^2(z_j)\,|\sin a_j| \prod_{i\in S_k\setminus\{j\}} |\cos a_i|$, and $\mathrm{sech}^2 \le 1$, $|\sin| \le 1$. Under independence the expectation of the squared product factorizes, and $\mathbb{E}[\pi^2\mathrm{sech}^4(z_j)\sin^2 a_j] \le \pi^2$. □

Under independence, each additional input in a readout's support therefore multiplies the bound on the mean squared gradient that every other input receives through that readout by $\mathbb{E}[\cos^2 a_i]$, which is below 1 unless a_i concentrates at 0 or $\pm\pi$; if $\rho = 1$ the bound need not fall. The proposition bounds the size of the per-readout gradient; it does not by itself show that the backbone learns a worse representation, and an input can still receive gradient through other readouts. On the trained bottlenecks (CIFAR-10, 4,000 images, seeds 0–2, validation embeddings), $\mathbb{E}[\cos^2 a_j]$ ranges from 0.37 to 1.0 across coordinates (mean 0.63 for the default circuit, 0.72 for the last-ring circuit), and the measured $\mathbb{E}[\prod_i \cos^2 a_i]$ follows the independence product closely: 0.39 vs 0.39 with two other factors for the default circuit, and 0.11 vs 0.09 with seven for the last-ring circuit, a factor of roughly 0.6–0.8 per additional input. Coordinates with $\mathbb{E}[\cos^2 a_j]$ near 1 sit at $a_j \approx \pm\pi$, where $\tanh$ saturates: they leave the other inputs' gradients unattenuated but receive almost none themselves.

E. Nonlinear Probe, Gradient Rank, Initialization and Gradient Decoupling

Table 9. Linear vs nonlinear probe on the same frozen bottlenecks (CIFAR-10, 4,000 training images; MLP probe: standardization, then $8 \to 256 \to 256 \to 10$ with ReLU, probe learning rate and epoch selected on validation; mean ± SD over n seeds; last column paired). The MLP probe recovers at most a fraction of a point more than the linear probe for any head, and VQC − linear is -4.8 ± 0.9 under the MLP probe vs -4.7 ± 0.9 under the linear probe.

Head the backbone was trained with	n	Linear probe	MLP probe	MLP − linear
linear	5	74.5 ± 0.5	74.4 ± 0.5	0.0 ± 0.1
MLP-8	5	70.8 ± 0.5	70.8 ± 0.6	0.0 ± 0.6
VQC, no CNOT	5	73.6 ± 0.5	73.6 ± 0.4	0.0 ± 0.4
VQC, pairs	5	72.9 ± 0.6	72.9 ± 0.5	0.0 ± 0.4
VQC, pairs, last	5	71.2 ± 0.7	71.2 ± 0.5	0.0 ± 0.4
VQC, ring (default)	5	69.8 ± 1.0	69.6 ± 0.8	-0.2 ± 0.3
VQC, ring, last	5	60.1 ± 3.1	60.3 ± 2.7	$+0.2 \pm 0.9$
cos	5	73.3 ± 0.4	73.2 ± 0.6	-0.2 ± 0.3
cos-prod. (default supports)	5	68.8 ± 1.6	69.0 ± 1.5	$+0.2 \pm 0.2$
cos-prod. (last-ring supports)	5	54.8 ± 3.4	55.2 ± 3.3	$+0.4 \pm 0.4$

Table 10. Gradient-decoupled classical product heads: same forward products, but each input's gradient passes only through its own cosine factor (straight-through on the other factors). CIFAR-10, 4,000 images, 3 seeds; decoupled heads at their validation-selected learning rate (grid 10^{-3}–3×10^{-2}), coupled heads as in Table 7. Decoupling fails to fit the training set, so it does not isolate the effect of coupling.

		Clean-train acc.		Probe	
Products	lr	coupled	decoupled	coupled	decoupled
default supports	0.003	84.0 ± 1.7	55.7 ± 3.7	67.8 ± 1.1	57.7 ± 1.5
last-ring supports	0.003	56.4 ± 5.2	42.4 ± 5.3	54.2 ± 3.1	48.9 ± 4.7

Table 9 refits the five-seed headline backbones of Figure 1b with a two-layer MLP probe under the same validation protocol as the linear probe; MNIST with 1,000 images gives the same picture (the MLP probe is within 0.5 points of the linear

Table 11. Effective rank (of 8) of the class signal the backbone receives: $\exp H(p)$, with p the normalized eigenvalues of $\frac{1}{N}\sum_i g_i g_i^\top$, $g_i = \partial \mathcal{L}_i / \partial z$ (cross-entropy) on the 2,000 validation embeddings (CIFAR-10, 4,000 training images; 5 seeds). Final: the trained run at its validation-selected lr. Init: a freshly initialized model of the same kind built with the run's seed. Δ: paired difference to the default circuit.

		Init		Final	
Head	Order	rank	Δ	rank	Δ
linear	—	5.4 ± 0.5	$+2.6 \pm 0.5$	7.3 ± 0.2	$+2.8 \pm 0.7$
VQC, no CNOT	1	4.0 ± 0.6	$+1.2 \pm 0.8$	7.2 ± 0.2	$+2.7 \pm 0.8$
VQC, ring (default)	2.5	2.8 ± 0.4	—	4.5 ± 0.6	—
VQC, ring, last	5.25	1.7 ± 0.3	-1.1 ± 0.4	3.7 ± 0.6	-0.8 ± 0.6

Table 12. Initialization control: probe accuracy (%) with the variational rotations initialized in $\mathcal{U}(-0.1, 0.1)$ (near the identity, the protocol default) or $\mathcal{U}(-\pi, \pi)$ (CIFAR-10, 4,000 training images; seeds 0–2; head lr selected on validation from $\{3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}\}$ at the large initialization). The last row is the paired difference between the two circuits.

Circuit	$\mathcal{U}(-0.1, 0.1)$	$\mathcal{U}(-\pi, \pi)$	large − small	selected lr
VQC, ring (default)	69.9 ± 1.1	69.0 ± 1.8	-0.9 ± 1.6	0.01 / 0.01
VQC, no CNOT	73.7 ± 0.6	73.6 ± 0.4	-0.1 ± 0.2	0.01 / 0.01
no CNOT − default	$+3.8 \pm 1.6$	$+4.6 \pm 1.7$		

probe for every head). With the original grid the MLP probe selected its lowest learning rate (10^{-3}) for the linear- and default-head backbones. Extending the grid down to 10^{-4} for the five headline heads (linear, default, ring-free, last ring, pairs; 5 seeds) changes every contrast against the default circuit by less than 0.1 points (VQC − linear -4.8 ± 0.7, every seed negative); one default-head probe still selects the new lowest rate. A stronger nonlinear decoder could still recover more; we claim only that the gap persists under these probes. Table 10 gives the decoupling attempt discussed in Section 6. Table 11 gives the effective rank of the backbone's loss gradient, and Table 12 the initialization control.

F. A Second Family of CNOT Layouts

Table 13. Out-of-sample test of product order on a second family of CNOT layouts (CIFAR-10, 4,000 images, small CNN, protocol of Section 3). The layouts were chosen by their zero-rotation product order alone, and their probe accuracy was predicted, before training, from a linear fit over the seven ring-family VQC circuits, seeds 0–2 (probe $= 76.7 - 3.22\times$ order). Spearman correlation between order and mean probe accuracy over the six layouts: -0.89; 5 of 6 within 4 points of the prediction. R^2: held-out fraction of the trained readouts' variance recovered by an affine map of the same circuit's zero-rotation readouts (Section B); its rank correlation with the prediction residual (observed − predicted, post hoc) is -0.94.

Layout	CNOTs	Order	Predicted	Probe	Obs. − pred.	R^2	n
star	$\mathrm{CNOT}(0, q)$, every layer	1	73.5	73.1 ± 1.2	-0.4	0.82	5
chain	$\mathrm{CNOT}(q, q{+}1)$, $q < 7$, every layer	1.5	71.9	71.4 ± 1.1	-0.5	0.84	5
star-in, last	$\mathrm{CNOT}(q, 0)$, last layer	1.875	70.7	73.0 ± 0.5	+2.4	0.70	5
ladder-2, last	$\mathrm{CNOT}(q, q{+}2)$, last layer	3	67.0	66.6 ± 0.5	-0.5	0.81	5
brickwork	alternating pairs, every layer	3.5	65.4	70.5 ± 0.6	+5.1	0.51	5
chain, last	open chain, last layer	4.5	62.2	58.2 ± 3.1	-4.0	0.89	5

Table 13 tests product order out of sample. The six layouts were chosen by their zero-rotation order alone, from CNOT patterns not used anywhere else in the paper, and their predicted probe accuracies and the pass criteria (Spearman correlation at most -0.8; at least four of six within 4 points) were recorded before any run. Every layout selected the head learning rate 10^{-2}, interior to its grid. Both criteria pass, but the order ranking is not fully preserved: two pairs reverse it in every seed (brickwork − distance-2 ring after the last layer $+4.0 \pm 0.7$; star-shaped ring after the last layer − open chain $+1.7 \pm 1.3$). The star-shaped ring gives one readout order 8 and the other seven order 1, and probes 2.4 points above the prediction from its mean order of 1.875; a single high-order readout may cost less than the mean suggests, a hypothesis we have not tested. Brickwork, which alternates its pairs between layers, probes 5.1 points above its prediction. Its supports are nearly uniform (every input in 3 or 4 of them), but so are those of the distance-2 ring after the last layer (2 to 4), which lands on its prediction, so support uniformity does not explain the miss. What does separate the two misses is how far training moves each circuit from its zero-rotation model (post hoc, Table 13): brickwork's trained readouts are the least well reproduced by

its products (held-out R^2 0.51, against 0.70–0.89 for the others; trained accuracy 70.2 vs 45.0 with the remapped products), it has the largest trained R_y rotations (mean $|\theta|$ 0.44), and the star-shaped ring has the next-lowest R^2 (0.70). Across the six layouts the prediction residual and R^2 have rank correlation -0.94. Order describes a circuit at its zero-rotation start, and layouts whose trained rotations carry them furthest from it escape part of the predicted cost; with six layouts this is an association, not a mechanism. Measured directly, brickwork's effective order at its trained rotations is 2.6 rather than 3.5 (Section G). The gradient effective rank of Table 11 (star 7.1, chain 5.7, star-shaped ring 6.5, distance-2 ring 4.6, brickwork 4.7, chain after the last layer 3.5) ranks the six layouts exactly as their probe accuracy does, including both reversals, but captures little of brickwork's advantage: its rank exceeds the distance-2 ring's by only 0.1 ± 0.6, against a 4-point probe gap. A two-layer MLP probe gives the same pattern (brickwork 70.2, open chain after the last layer 58.8).

G. Effective Order Beyond Zero Rotations

Table 14. Effective order beyond zero rotations (CIFAR-10, 4,000 images, small CNN; *: second family). For rotations θ, readout k is $\langle\psi(z)|O_k|\psi(z)\rangle$ with $O_k = U(\theta)^\dagger Z_k U(\theta) = \sum_P c_P P$ over Pauli strings; strings with a Y factor vanish on the encoded state, and an X or Z factor on qubit j makes the readout depend on input j. The effective order is the c_P^2-weighted mean number of inputs a visible string touches, averaged over readouts; at $\theta = 0$ it is the product order of Equation (1). Trained: the trained rotations (5 seeds; 3 for the 1- and 2-layer rings). Random: $\theta \sim \mathcal{U}(-\pi, \pi)$ (20 draws), with the share of weight on invisible (Y) strings. Spearman correlation with probe accuracy over the 13 layouts: -0.91 at $\theta = 0$, -0.94 trained (post hoc), -0.71 random.

	Effective order				
Layout	$\theta = 0$	trained	random	Y share (random)	Probe
no CNOT	1.00	1.00	1.00	0.33	73.6
pairs	1.00	1.06	1.46	0.45	72.9
pairs, last	1.50	1.50	1.50	0.42	71.2
ring (default)	2.50	2.65	5.44	0.90	69.8
ring, 2 layers	5.25	5.12	5.39	0.80	61.2
ring, 1 layer	5.25	5.25	5.25	0.00	56.8
ring, last	5.25	5.25	5.25	0.85	60.1
star*	1.00	1.01	3.41	0.75	73.1
chain*	1.50	1.55	4.16	0.84	71.4
star-in, last*	1.88	1.88	1.88	0.40	73.0
ladder-2, last*	3.00	3.00	3.00	0.72	66.6
brickwork*	3.50	2.62	2.88	0.68	70.5
chain, last*	4.50	4.50	4.50	0.78	58.2

Product order is defined at zero rotations. Table 14 extends it to any rotations by expanding each readout operator in Pauli strings and counting, with weight c_P^2, the inputs that each string visible on the encoded state touches; the test suite checks that the expansion reproduces the simulator's readouts and input dependences exactly, and that it equals Equation (1) at $\theta = 0$. Three observations follow. First, layouts with a single CNOT layer after the last rotations cannot change their effective order at any rotations, because single-qubit rotations do not change which qubits a string acts on; only every-layer layouts can move, and of these brickwork moves most, from 3.5 to 2.6 at its trained rotations, below the distance-2 ring's 3.0, which restores that pair's probe ranking and shrinks its residual from the fit from $+5.1$ to $+2.3$ points. Second, at random rotations $\theta \sim \mathcal{U}(-\pi, \pi)$ the default ring's readouts depend on all eight inputs (effective order 5.4) while the ring-free ones stay at one, consistent with the gap at large initialization (Table 12). Third, random-rotation order predicts probe accuracy worse than order at $\theta = 0$ (Spearman -0.71 vs -0.91): the star and the open chain reach 3.4 and 4.2 at random rotations yet probe near order-1 circuits, because trained circuits stay close to their zero-rotation regime. Order computed at the trained rotations ranks the circuits slightly better (-0.94), but it uses the trained weights and is therefore a post hoc description, not a prediction.

H. Pretrained-Backbone Controls

Table 15 separates training size from the number of optimizer updates on the pretrained ResNet-18, and Table 16 gives the lockstep comparison of the trained circuit, the circuit frozen at zero rotations and the products head discussed in Section 5.6. A step-by-step check (seed 0, head lr 10^{-3}) shows why the lockstep pairs still differ: two copies of the products head trained on the same batches stay bit-identical, while the frozen circuit and the products head, whose outputs agree to floating-point rounding (a test checks 10^{-5}), reach a backbone parameter distance of 2×10^{-5} after one update, 0.14 after 21 and 3.7

after 125 (one epoch), with the backbone's parameter norm at 85.

Table 15. Training budget on the pretrained ResNet-18 (CIFAR-10; seeds 0–2; probe accuracy, %). The protocol trains 30 epochs at every size, so smaller training sets get fewer optimizer updates. Matched rows train 120 epochs at 1,000 images and 12 at 10,000, approximately the update count of 4,000 images × 30 epochs (see the updates column), with the head learning rates selected at 30 epochs for each size or re-selected on validation over a half-decade step either way (re-tuned). Differences are paired over seeds, against the linear head.

Images	Epochs	Updates	linear	default − linear	no CNOT − linear
1,000	30	960	82.9 ± 0.9	-22.3 ± 2.9	-1.6 ± 1.2
1,000	120 (matched)	3,840	81.7 ± 0.9	-6.6 ± 1.7	$+0.2 \pm 1.3$
1,000	120 (matched, re-tuned)	3,840	83.3 ± 0.6	-8.3 ± 2.1	-1.4 ± 1.2
4,000	30	3,750	89.5 ± 0.3	-5.7 ± 0.7	0.0 ± 0.2
10,000	30	9,390	92.6 ± 0.4	-1.9 ± 0.4	-0.1 ± 0.4
10,000	12 (matched)	3,756	91.4 ± 0.3	-7.2 ± 2.1	$+0.5 \pm 0.2$
10,000	12 (matched, re-tuned)	3,756	92.6 ± 0.1	-7.7 ± 1.9	-0.7 ± 0.2

Table 16. Lockstep control on the pretrained ResNet-18 (CIFAR-10, 4,000 images). The trained default circuit, the same circuit with its rotations frozen at exactly zero (forward map and input gradients equal to the products head's), and the classical head on the default circuit's zero-rotation products are trained in one process from copied initial weights on identical batches; paired probe differences over seeds 0–9 at each head lr, with a 95% paired t interval. The last row pairs the heads at the rates validation selected for them in the main grid. Mean probe accuracy (trained / frozen / products), 3×10^{-4}: 82.0 / 82.7 / 82.6; 10^{-3}: 85.6 / 83.9 / 84.9.

Paired difference	mean ± SD	95% interval	negative
trained − products, both at 3×10^{-4}	-0.6 ± 2.0	[-2.0, +0.8]	5/10
frozen − products, both at 3×10^{-4}	0.0 ± 0.6	[-0.4, +0.5]	4/10
trained − products, both at 10^{-3}	$+0.6 \pm 3.9$	[-2.1, +3.4]	5/10
frozen − products, both at 10^{-3}	-1.0 ± 2.6	[-2.9, +0.8]	6/10
trained at 3×10^{-4} − products at 10^{-3} (main-grid rates)	-2.9 ± 3.0	[-5.1, -0.7]	8/10

I. Exploratory Sweep with One Shared Recipe

Before the protocol of Section 3, we ran every head with one shared recipe: head learning rate 10^{-3} (backbone 5×10^{-4}), 25 epochs on MNIST and 80 on CIFAR-10, three data re-uploading schedules for the VQC, and checkpoints chosen by best test-split accuracy. Because the test split took part in checkpoint selection, these numbers are exploratory; mean final-epoch accuracy is within 1.35 points of best-epoch accuracy in all 30 configurations and no ordering of heads changes (`scripts/make_tables.py` on the released per-epoch logs in `results/raw`). Tables 17 and 18 give the results. Re-uploading lowered accuracy in every setting under this recipe (NONE > LIGHT > STRONG); it was not part of the tuned protocol.

Table 17. Exploratory sweep, MNIST: best test-split accuracy (%, mean ± SD over 3 seeds) with one shared recipe. Bold marks the better of MLP-8 and the VQC (NONE).

			VQC (re-uploading schedule)		
N_{train}	Linear	MLP-8	NONE	LIGHT	STRONG
250	61.5 ± 1.7	36.7 ± 5.4	$\mathbf{54.5 \pm 3.7}$	48.3 ± 10.7	38.3 ± 7.4
500	79.3 ± 2.9	53.2 ± 6.9	$\mathbf{71.2 \pm 1.9}$	57.8 ± 2.5	45.6 ± 6.7
1,000	89.6 ± 0.7	76.4 ± 3.4	$\mathbf{83.3 \pm 0.3}$	68.4 ± 3.5	54.6 ± 14.0

J. Frozen-Embedding Transfer

To separate a head from the backbone it induces, we also trained fresh heads on fixed embeddings extracted from exploratory-sweep backbones (linear- or VQC-trained; MNIST at 1,000 images, CIFAR-10 at 4,000; 3 seeds; 498 head fits with learning rate and epoch selected on validation). Table 19 gives the results. On VQC-trained embeddings, a refitted linear head beats a refitted VQC by 1.7 ± 0.8 (MNIST) and 1.6 ± 0.5 points (CIFAR-10), positive for every seed. On linear-trained embeddings, most of whose values lie in the flat region $|\tanh z| > 0.9$ of the encoding (56% and 66%), the VQC collapses, trailing

Table 18. Exploratory sweep, CIFAR-10: best test-split accuracy (%, mean ± SD over 3 seeds) with one shared recipe. Bold marks the better of MLP-8 and the VQC (NONE).

			VQC (re-uploading schedule)		
N_{train}	Linear	MLP-8	NONE	LIGHT	STRONG
1,000	57.5 ± 0.8	$\mathbf{51.2 \pm 0.8}$	37.1 ± 2.4	34.2 ± 0.7	26.6 ± 1.9
2,000	64.6 ± 0.7	$\mathbf{60.6 \pm 0.9}$	46.8 ± 5.2	42.7 ± 3.9	28.7 ± 1.9
4,000	71.2 ± 0.5	$\mathbf{67.2 \pm 0.4}$	58.9 ± 4.6	52.0 ± 7.8	35.2 ± 5.5

the linear head by 20.9 ± 4.4 and 41.8 ± 2.9 points, and a classical Fourier head $[\cos a_j, \sin a_j] \to$ Linear with the same encoding collapses almost as far on CIFAR-10. Fixed standardization of the embedding recovers 12.6 ± 6.2 and 30.4 ± 2.9 of those points for the VQC, and a learned per-dimension affine 15.0 ± 7.6 and 35.9 ± 0.9; tripling the training budget recovers 0.5 ± 0.2 on CIFAR-10. The Fourier head's recovery under standardization was not tested. The source checkpoints were selected on the test split, so this experiment is exploratory; it motivated the probe measurements of Section 5, which use validation-selected backbones only.

Table 19. Frozen-embedding transfer: test accuracy (%, mean ± SD over 3 seeds) of heads trained on identical frozen embeddings. Columns give the head the backbone was trained with and the fraction of test embedding values with $|\tanh z| > 0.9$. †Selected learning rate is the largest tried for that head.

	MNIST (1,000 images)		CIFAR-10 (4,000 images)	
Head	linear src. (56%)	VQC src. (5%)	linear src. (66%)	VQC src. (30%)
Linear	89.8 ± 0.2	88.9 ± 1.6	70.9 ± 0.6	$62.5 \pm 4.4^{\dagger}$
MLP-8	88.3 ± 0.9	88.5 ± 1.1	70.2 ± 0.8	62.3 ± 4.0
Fourier	78.2 ± 1.1	87.1 ± 2.0	33.4 ± 1.0	61.8 ± 4.5
VQC	69.0 ± 4.4	87.2 ± 1.0	29.1 ± 3.0	61.0 ± 4.1
VQC+std	81.6 ± 5.6	77.6 ± 6.6	59.5 ± 0.4	53.7 ± 5.6
VQC+std+affine	84.0 ± 3.3	82.7 ± 1.6	65.0 ± 3.1	61.6 ± 4.4
VQC, 3× epochs	–	–	$29.6 \pm 3.0^{\dagger}$	$61.0 \pm 4.5^{\dagger}$

K. Learning-Rate Sensitivity

Trigonometric heads, quantum or classical, tolerate a narrow range of head learning rates under the coupled recipe (backbone at half the head rate). On CIFAR-10/4,000 all of them select 10^{-2}, and at 3×10^{-2} test accuracy falls to 63.4 for the default VQC, 50.0 for the ring-free VQC, and 29.3 and 28.9 for the classical cos and cos+sin heads, while the linear head selects 3×10^{-2}. With the head learning rate fixed, the default VQC also collapses at backbone learning rates of 3×10^{-2} and 6×10^{-2}, where its bottleneck saturates the encoding almost completely, whereas the linear head still trains (Table 3). Test accuracy for every grid point is in the released results.

L. Sign Symmetry

At zero rotations every readout is an even function of each z_j, because cosine is even. We measured how far trained heads are from this symmetry on CIFAR-10/4,000 test embeddings: the median, over samples, of $\|q(-\mathbf{z}) - q(\mathbf{z})\| / \|q(\mathbf{z}) - q(\mathbf{z}')\|$, with q the readouts and $\mathbf{z}'$ a random other sample (0: blind to the sign; about 1: flipping the sign moves the readout as far as changing the image). The frozen circuit stays nearly sign-blind (0.09–0.12), and its accuracy on $-\mathbf{z}$ is within 1.4 points of that on $\mathbf{z}$. Trained heads are not: 0.58–0.73 for the default VQC (accuracy on $-\mathbf{z}$: 25–34%), 0.91–1.02 for the ring-free VQC (6–17%), and 0.85–0.97 for the monotone and symmetric encodings of Table 3. Sign-blindness at initialization therefore does not predict the backbone cost, consistent with the null result of the monotone encoding.

M. Training and Reproducibility Details

Training. AdamW with weight decay 10^{-4}, cosine annealing to zero, batch size 32, gradient clipping at norm 1.0 (MNIST) or 2.0 (CIFAR-10); CIFAR-10 augmentation is random crop, horizontal flip, small rotation and color jitter. Head learning-rate grids: MNIST $\{3 \times 10^{-4}, \ldots, 3 \times 10^{-2}\}$ and CIFAR-10 $\{10^{-3}, 3 \times 10^{-3}, 10^{-2}\}$ in half-decade steps, each

extended by one step whenever the selected rate was at the top; circuit variants $\{3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}\}$; classical product heads $\{10^{-3}, \ldots, 3 \times 10^{-2}\}$ on CIFAR-10 circuit supports and $\{3 \times 10^{-3}, \ldots, 3 \times 10^{-2}\}$ on designed supports and on MNIST (extended to 10^{-1} for the MNIST last-ring products). All selected rates of circuit and classical product heads are interior to their grids, except in the re-tuned matched-budget search on the pretrained backbone (one half-decade step either way), where the 10,000-image default head and the linear head at both sizes select a boundary rate; that search is local. The probe uses AdamW (weight decay 10^{-4}), batch size 128, 60 epochs with a cosine schedule, and probe learning rates $\{3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}, 10^{-1}, 3 \times 10^{-1}, 1\}$, all selected on validation.

Reproducibility. The repository contains per-epoch metrics and metadata for every training run, the probe, frozen-head and trained-circuit measurements, and the scripts that regenerate every number, table and figure from them (`scripts/summarize_v2.py`, `scripts/zero_angle_check.py`, `paper/figures/scripts/`). The PyTorch statevector simulator is tested against PennyLane (tolerance 10^{-6} in outputs and input gradients, 10^{-5} in rotation gradients); the test suite also checks the zero-rotation identity, the ring-free form, the entangling example of Section A, the equality of the zero-rotation circuit and the products head, and the effective-order expansion of Section G against the simulator. The predicted orderings for the MNIST replication and the predictions and pass criteria for the second family of layouts were written in the project's log before those runs. Seeds control data subsets, initialization and data order. Experiments ran on one Apple M3 Max.